

MINI ASSIGNMENT

Deconstructing HCI & Computer Graphics in Daily Software

Course Name — Introductory HCI & Computer Graphics

NAME

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ROLL NUMBER

2k24/CSE/62

DEPARTMENT

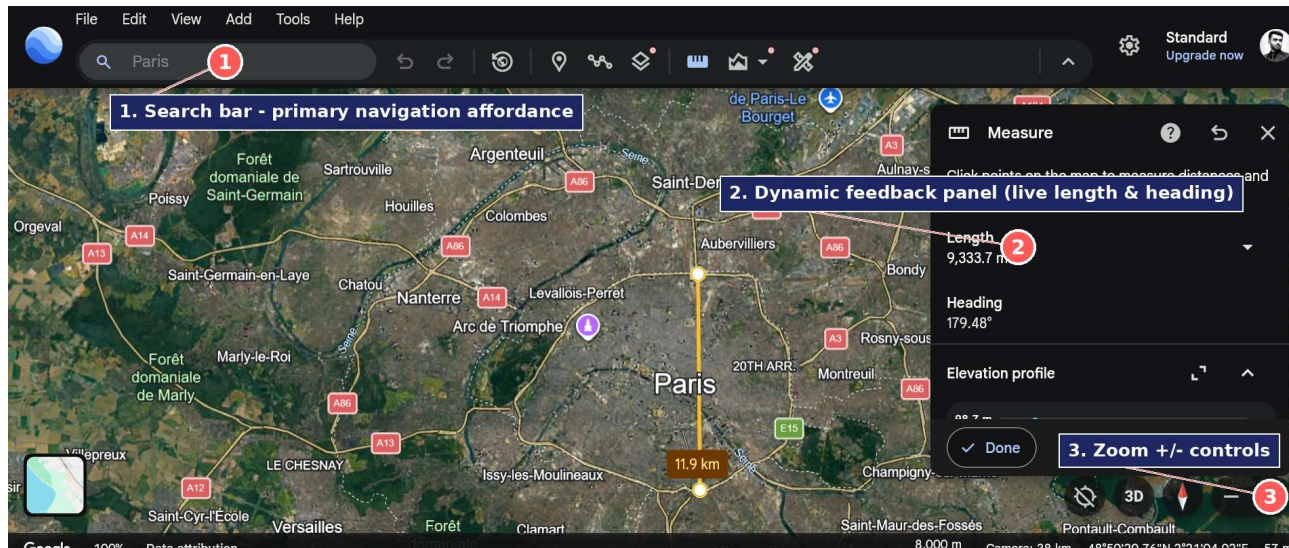
IMCS – Computer Science PE

TEACHER'S NAME

Sir Rajeesh

Human-Computer Interaction (HCI) Focus

App: Google Earth | View: Measure tool over the Paris region



1. Search bar

A single, prominent text field lets users jump to any location instantly — a clear affordance that invites typing without instructions.

2. Dynamic feedback panel

The Measure tool updates Length and Heading live as the user clicks points, giving immediate, continuous feedback on the action performed.

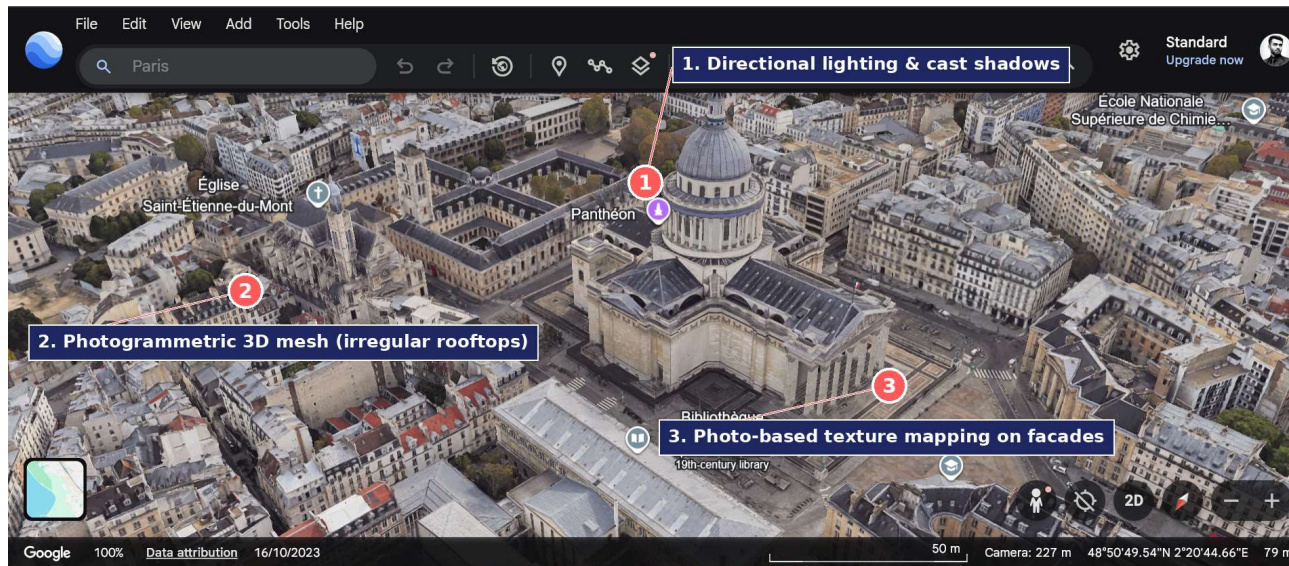
3. Zoom +/- controls

Simple, familiar plus/minus buttons give users direct manipulation over scale, reinforcing recognition over recall.

Why this matters for HCI: Each of these elements reduces the user's cognitive load — the search bar removes the need to manually pan/zoom to a place, the feedback panel confirms the result of an action in real time, and the zoom controls map a familiar physical gesture (zooming in/out) onto simple, discoverable buttons.

Computer Graphics (CG) Focus

App: Google Earth | View: 3D photorealistic view of the Panthéon, Paris



1. Directional lighting & shadows

Sunlight casts consistent shadows off the dome and rooftops, giving strong depth cues that make the flat image read as a 3D volume.

2. Photogrammetric 3D mesh

Building rooftops and edges are irregular and detailed rather than flat polygons, showing the underlying 3D mesh reconstructed from real aerial imagery.

3. Photo-based texture mapping

Real photographic textures are mapped onto the building facades and streets, giving surfaces realistic color and detail instead of plain shading.

Reflection

1. How does CG quality affect interaction? Google Earth's realistic lighting and photogrammetric 3D view help the user quickly orient themselves and judge scale and distance, but the heavy detail can also slow rendering and make precise clicking (e.g., placing a measure point) harder on dense rooftops.

2. One HCI improvement: Add a visible on-screen legend or tooltip explaining each toolbar icon on hover, since several icons (measure, layers, Street View) currently rely on the user already knowing what they do.